

Impact of Vector Databases on Information Retrieval for Artificial Intelligence Agents Specialized in SDN API Documentation

A Comparative Study of FAISS, PGVector, and Neo4j in
Retrieval-Augmented Generation Architectures

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Abstract

As modern networks become increasingly software-defined and API-driven, the ability of automation systems to understand and leverage their own capabilities has become a critical requirement. Network automation platforms often expose hundreds of APIs that enable configuration, monitoring, troubleshooting, and operational management tasks. However, these capabilities are typically documented in large and complex technical repositories, making it difficult for both automation systems and network engineers to efficiently discover and utilize the appropriate APIs for a given task. As a result, the full potential of network automation is often limited by the accessibility and usability of the available documentation.

Retrieval-Augmented Generation (RAG) architectures offer a promising approach for addressing this challenge by enabling Artificial Intelligence (AI) agents to retrieve and reason over technical documentation in real time. Such agents could serve as an intelligent knowledge layer that allows automation systems to become aware of their available capabilities, dynamically identify suitable APIs, and support autonomous configuration, troubleshooting, and operational decision-making. This study investigates the impact of different vector database technologies on the effectiveness of this approach by comparing FAISS, PGVector, and Neo4j within a RAG-based architecture focused on SDN API documentation. The evaluation considers information retrieval accuracy, response quality, and retrieval latency to determine how the choice of vector database influences the performance of AI agents designed to enhance network automation and assist engineers in accessing complex technical information. Through this comparison, the research aims to identify the most suitable retrieval architecture for enabling intelligent, self-aware automation systems in Software-Defined Networking environments.